DEEP LEARNING PROGRAMMING FINAL PROJECT

D3Net-based All-in-One Restoration for Smartphone Images with Dirty Lenses

SIDL adaptation, paper-baseline comparison, continued training, and failure analysis

20223081 신준철 · Deep Learning Programming

DATASET

Problem Setup

- SIDL provides real-world degraded-clean pairs captured with contaminated smartphone lenses.
- Lens contamination causes local artifacts, contrast loss, haze, and frequency-domain distortion.
- This project analyzes D3Net adaptation under a limited class-project training budget.

SIDL validation axis

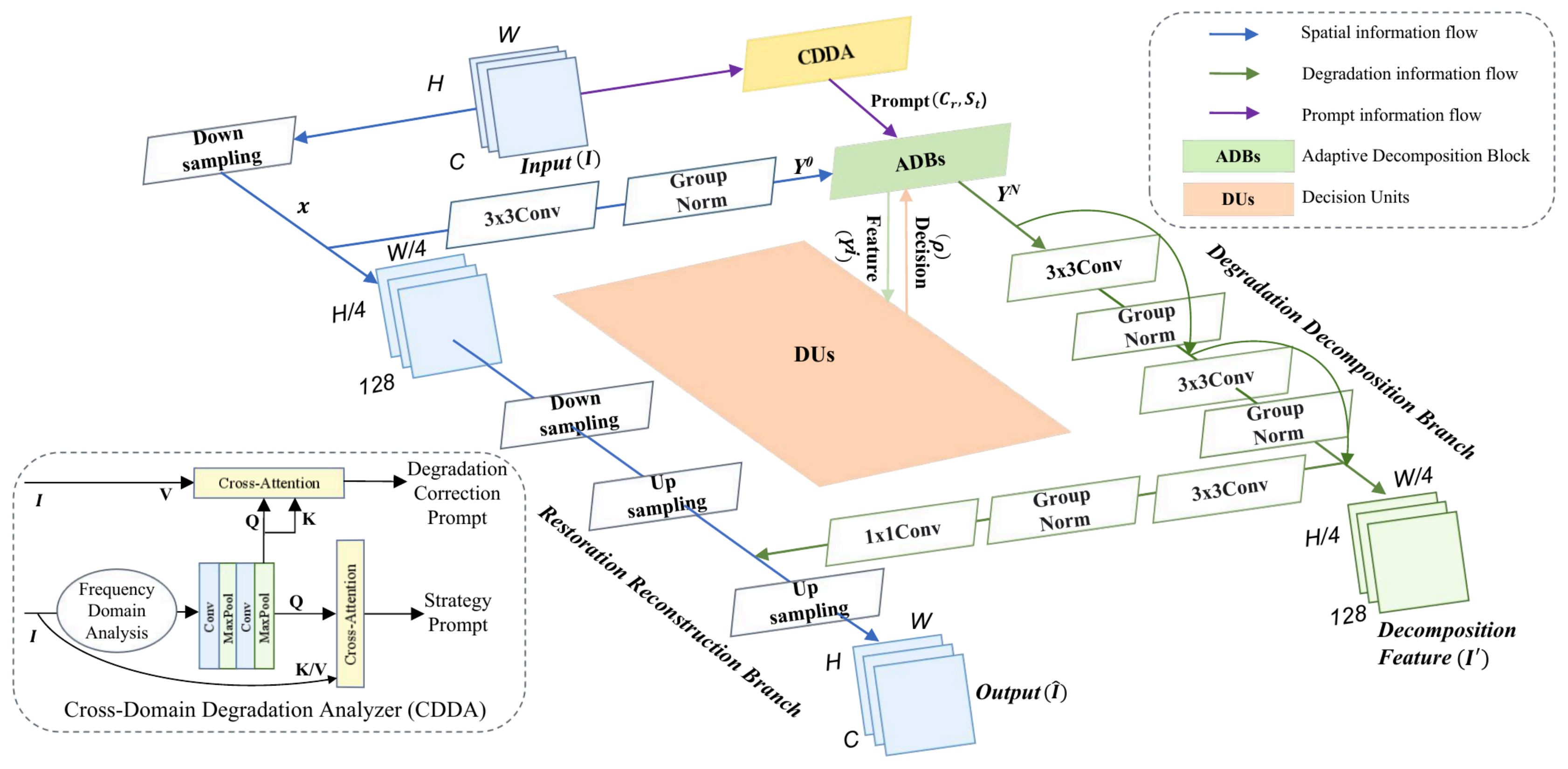
512×512 patchified image pairs

Clean / Dust / Finger / Water / Scratch / Mixed

Easy / Medium / Hard difficulty

Metrics: PSNR / SSIM

D3Net Architecture



Core components

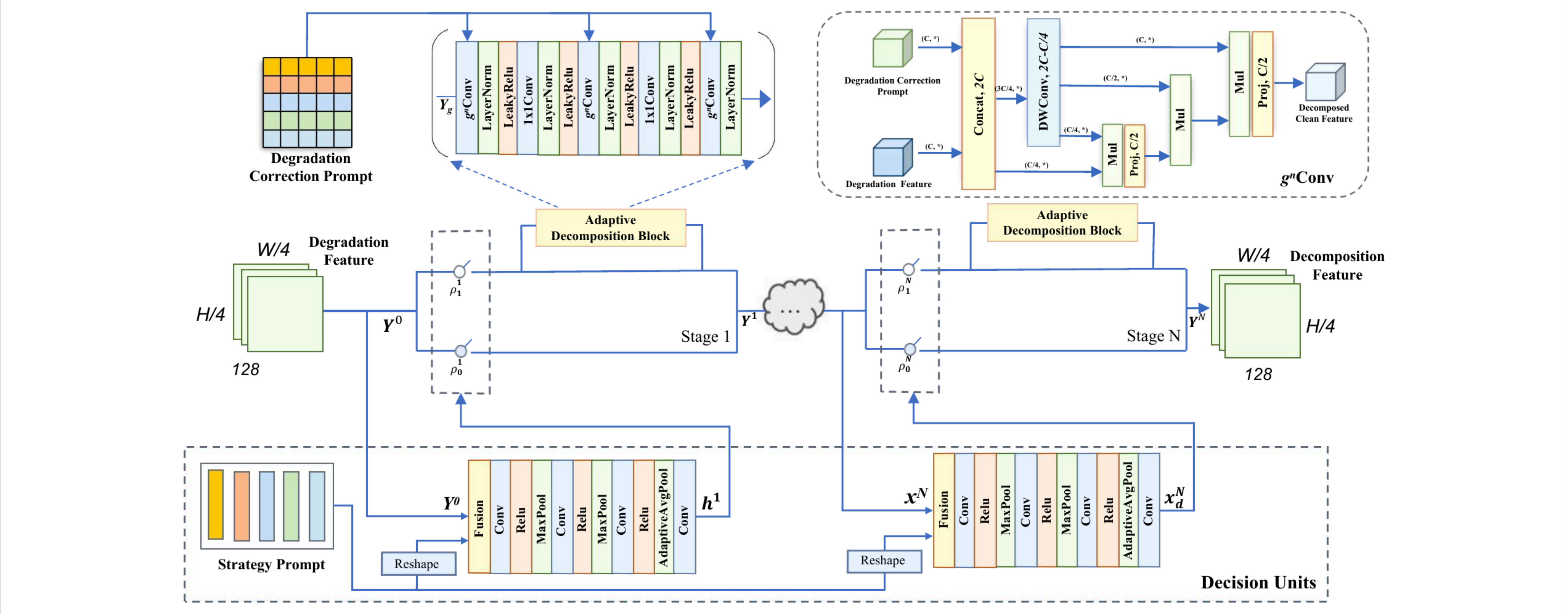
Residual prediction: restored image = input + residual

Fourier-domain cues for degradation representation

Dynamic degradation decomposition for mixed contamination

The model is used as an all-in-one backbone rather than a degradation-specific model.

Dynamic Decomposition



D3Net uses prompt-conditioned feature processing so that local and mixed contamination can be handled within one restoration model.

SETUP

Experimental Protocol

Main setting

256×256 random crop, scratch training, batch 8

Validation

512×512 center crop, image-wise PSNR / SSIM

Optimization

L1 loss, Adam, cosine LR, single RTX 4090

D3Net original

Official setting is roughly 1200 epochs

Official submission

Epoch 73, 771 GMACs, 43.8M parameters

Evaluation

Residual output, denormalize, clamp to [0,1], then compute PSNR/SSIM

BASELINE COMPARISON

Baseline Comparison

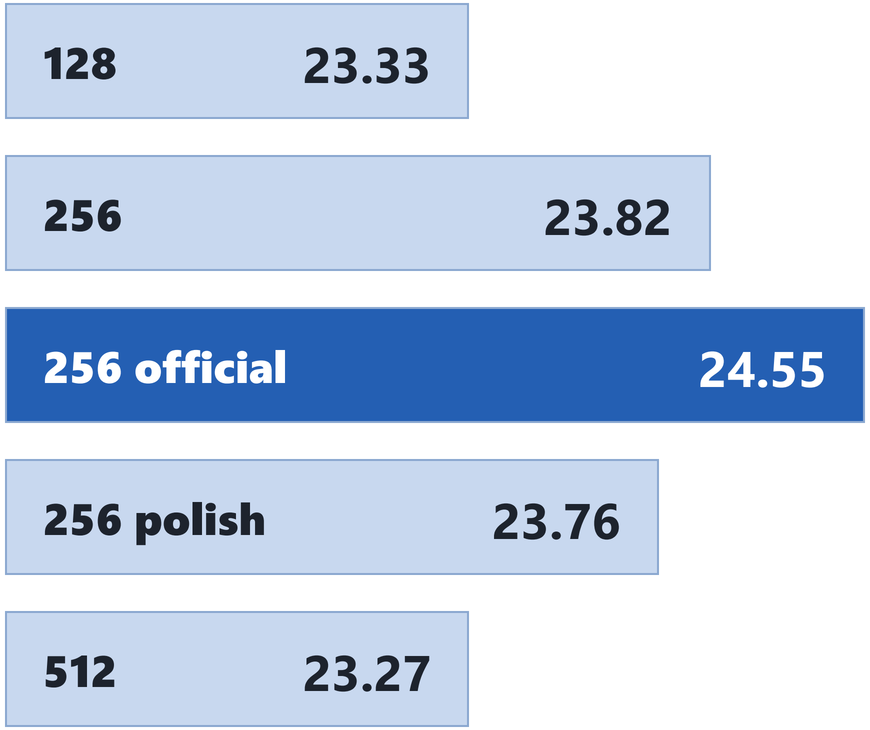
METHOD	LEVEL	CLEAN	DUST	FINGER	WATER	SCRATCH	MIXED	AVG.
AirNet	**Easy**	27.10	23.40	25.10	24.85	25.84	25.30	25.26
	Medium	26.13	19.69	22.53	20.36	21.24	16.79	21.12
	Hard	23.79	16.18	15.83	16.64	18.13	14.48	17.50
DiffUIR	**Easy**	34.51	26.96	28.82	27.42	30.14	28.22	29.34
	Medium	33.25	22.11	26.16	24.65	27.07	20.32	25.36
	Hard	29.47	20.38	18.93	19.91	21.27	18.97	21.71
D3Net ours	**Easy**	**33.20**	**25.10**	**26.95**	**26.45**	**28.84**	**27.56**	**28.02**
	Medium	**30.57**	**22.53**	**25.50**	**23.73**	**25.69**	**20.18**	**24.70**
	Hard	**27.63**	**20.21**	**18.09**	**19.47**	**21.32**	**18.82**	**20.92**

AirNet/DiffUIR are SIDL paper test-set values. D3Net is our official leaderboard submission using the epoch-73 checkpoint.

RESULTS

Ablation Study

RUN	INIT.	CROP	EPOCH	PSNR	SSIM
128 baseline	scratch	128	34	23.33	0.8357
256 main	scratch	256	34	23.82	0.8398
256 official	**latest**	**256**	**73**	**24.55**	**0.8507**
256 low-LR	256 best	256	3	23.76	0.8409
512 refinement	256 best	512	2	23.27	0.8336



INTERPRETATION

Official Test Breakdown

Official leaderboard average: 24.55 dB / 0.8507.

Easy average: 28.02 dB / 0.9051.

Hard average: 20.92 dB / 0.8064.

Key limitation

The original D3Net training scale is roughly 1200 epochs. Our submitted checkpoint is epoch 73, so the result should be interpreted as limited-budget adaptation.

The training loss is L1 on normalized tensors, while evaluation uses PSNR/SSIM on denormalized [0,1] images. This loss-metric mismatch may affect convergence.

FAILURE ANALYSIS

Failure Analysis

TYPE	AVG. PSNR	AVG. SSIM
Clean	30.47	0.9256
Dust	22.61	0.8312
Finger	23.51	0.8163
Water	23.22	0.8357
Scratch	25.28	0.8695
Mixed	22.19	0.8261

Hard Finger **18.09 / 0.6940**

Hard Mixed **18.82 / 0.8089**

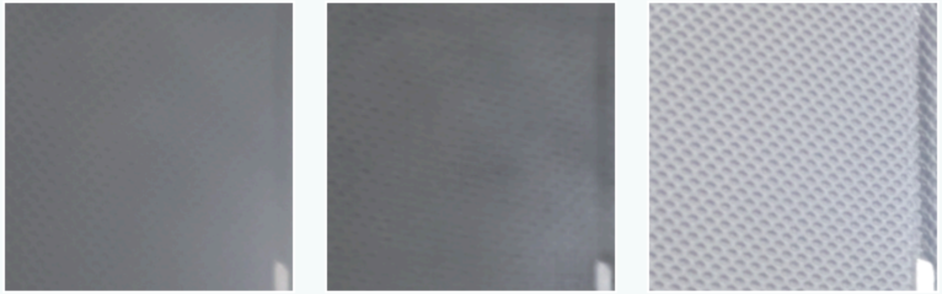
Severe local residue and ambiguous mixed contamination remain difficult to separate from true image structure.

Qualitative Results

1. finger / hard / Case051_F_001.png PSNR 19.28 dB SSIM 0.9316

2. dust / medium / Case251_D_018.png PSNR 19.25 dB SSIM 0.7943

3. scratch / hard / Case073_S_001.png PSNR 10.47 dB SSIM 0.6080



4. water / hard / Case051_W_001.png PSNR 17.70 dB SSIM 0.8999

Each panel shows input / D3Net output / GT using original report assets.

CONCLUSION

Summary & Future Work

Main result

Official test result: 24.55 dB / 0.8507

Key finding

Performance is strong on Clean/Easy, but Hard Finger and Mixed remain difficult.

Limitation

The epoch budget is far smaller than the original D3Net training scale.

Future work

Long training, pretrained D3Net, 512 curriculum, Charbonnier/L1+MSE losses, and TTA